# Hisu Kim

tomm1203@gm.gist.ac.kr | Google Scholar | hiskim1.github.io

## Education

**Gwangju Institute of Science and Technology, South Korea** — Sep. 2025 – Aug. 2027 (expected)
*M.S. in Climate Analysis and Modeling Lab (Advisor: Jin-ho Yoon)* — TGPA 4.37/4.5

**Gwangju Institute of Science and Technology, South Korea** — 2019 – 2025
*B.S. in Electrical Engineering and Computer Science* — TGPA 4.10/4.5
*and Earth Science and Environmental Engineering*
*Minor in Mathematics*

**Warsaw University of Technology, Poland** — Oct. 2022 – Feb. 2023
*Exchange student, Faculty of Building Services, Hydro and Environmental Engineering* — TGPA 4.86/5.0

**University of California, Berkeley, USA** — June – Aug. 2023
*Summer Session C* — TGPA 4.0/4.0

## Publications and Conferences

### Publications

- **Kim, H.**, Ryu, J., Son, S., Jeong, J., Kim, H., & Yoon, J.-H. (2026). A spectral test of the butterfly effect and physical consistency in the diffusion-based GenCast's ensembles. *npj Climate and Atmospheric Science*. https://doi.org/10.1038/s41612-026-01380-1
- Ryu, J., **Kim, H.**, Wang, S.-Y., & Yoon, J.-H. (2026). Increasing resolution and accuracy in sub-seasonal forecasting through 3D U-Net: The western US. *Geoscientific Model Development 19*(1), 27–39. https://doi.org/10.5194/gmd-19-27-2026

### Conferences

- **Kim, H.**, Ryu, J., Jang, W., Kwon, M., & Yoon, J. (2026, September). Uncertainty Assessment in Ensemble Forecast Post-processing Using U-Net and Diffusion. *Accepted at S2S2D 2026*, Reading, United Kingdom
- **Kim, H.**, & Yoon, J.-H. (2025, December 16). Revealing the butterfly effect in deep learning weather prediction model via kinetic energy spectra change. *Oral presentation at AGU25 Meeting*, New Orleans, LA, United States.

## Research and Work Experience

**Graduate Researcher** — Sep. 2025 – Present
*Climate Analysis and Modeling Lab, GIST* — *Advisor: Prof. Jin-ho Yoon*

- **GraphFourierCast**
  : Developing a Graph Fourier Transform framework utilizing Dirichlet energy of icosahedral mesh graph to diagnose latent-space smoothing in graph neural network-based weather models (ongoing research)
- **Team Lead, "CAMExpedition" (ECMWF AI Weather Quest 2025)**
  : Leading our team's entry in ECMWF's international AI/ML sub-seasonal forecasting competition; implementing the 4-source Bayesian uncertainty quantification framework of van Leeuwen et al. (2025) atop IFS ensembles with a geo-aware 3D U-Net extended by FiLM lead-time conditioning and attention-gated skip connections (ongoing research / accepted at S2S2D 2026)
- **Spectral test of the butterfly effect in GenCast**
  : Diagnosed the physical consistency of the diffusion-based GenCast ensembles via kinetic-energy (KE) and difference-KE spectra in the spherical-harmonic domain, with Helmholtz decomposition and $|\nabla \mathrm{KE}|$ gradient analysis (*npj Clim. Atmos. Sci.*, 2026)

**Undergraduate Research Intern** — Feb. 2023 – Aug. 2025
*Climate Analysis and Modeling Lab, GIST* — *Advisor: Prof. Jin-ho Yoon*

- Investigated perturbation responses in GraphCast to characterize the data-driven analog of atmospheric butterfly effects, laying the groundwork for the subsequent spectral analysis published in *npj Clim. Atmos. Sci.* (2026)

**Cybersecurity Analyst** Feb. 2021 – Oct. 2022
*Republic of Korea Naval Fleet Command* *Computer Emergency Response Team*

- Co-administered dozens of Linux production servers across the Fleet Command network
- Mitigated *log4j* and *MS-MSDT* vulnerabilities via NAC-distributed registry patches and removal of exploitable files; analyzed firewall/IPS packets and logs to flag unauthorized or anomalous communications

## Teaching Experience

**Calculus (Single- & Multivariable)** | *Teaching Assistant* 2025 Spring – 2026 Spring

- Single-variable (2026 Spring): designed weekly problem sets, prepared and led recitation sessions, and provided written feedback on students' solutions
- Multivariable (2025 Spring & Fall): held weekly office hours, answering conceptual questions and guiding students through theorem proofs

**Graph Theory** | *Teaching Assistant* 2024 Fall

- Conducted weekly office hours, answering questions and guiding students through theorem proofs
- Organized regular sessions to provide detailed feedback on homework and quizzes, and advanced mathematical LaTeX usage
- Offered supplementary lessons on challenging topics and extra problems

**Literature Courses from Prof. Soo-Jeong Lee** | *Teaching Assistant* 2023 Spring – 2026 Spring

- Built syllabus generator website to streamline the professor's course preparation process
- Used LaTeX to edit e-books compiled from students' story-writing assignments

**Digital Design** | *Teaching Assistant* 2023 Spring

- Conducted office hours twice a week answering questions from homework
- Created sample exam questions to facilitate student learning

## Professional Development

**Rossbypalooza 2026** 20 – 31 July 2026

- University of Chicago, IL, USA

**Advanced School on Foundation Models for Scientific Discovery** 10 – 18 July 2025

- International Centre for Theoretical Physics, Italy (virtual)

**Summer School of Mathematics: Algorithms** 23 – 27 June 2025

- Eötvös Loránd University, Hungary

## Scholarships and Fundings

**Presidential Science Scholarship for Graduate Student** | *Ministry of Science and ICT* 2026-7 (Accepted)
**Government Funded Scholarship for GIST Student** | *GIST* 2019 Spring – Present
**Scholarship for Outbound Program** | *GIST* 2023 Fall
**Academic Excellence Scholarship** | *GIST* 4 Semesters
**Scholarship for Summer Session Program Abroad** | *GIST* 2023 Summer

## Skills

| | | |
|---|---|---|
| Programming | Python | (Main) |
| | C/C++, C#, JavaScript, Bash | (Intermediate) |
| Libraries | PyTorch (Lightning), JAX, xarray, Dask, zarr | |
| Tools | Linux, Git, LaTeX, NCL, GrADS | |
| Language | Korean (native) | |
| | English (fluent) | |
| | Spanish, Polish (elementary) | |